

SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE - 3

Name : HARSH RAJ SHARMA

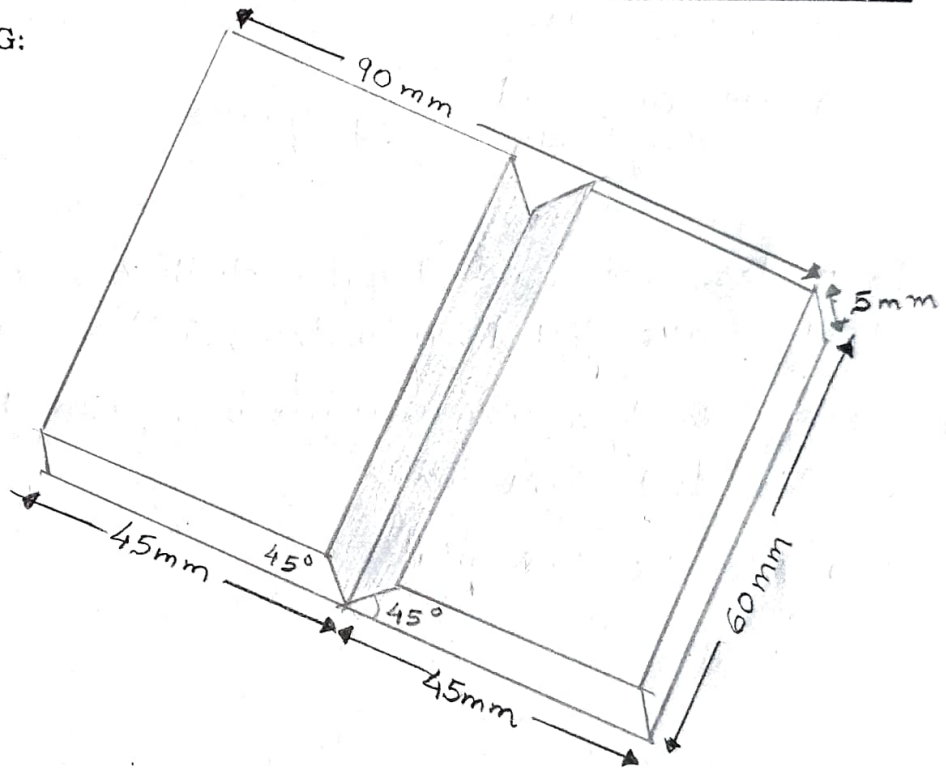
WELDING - SHOP

Roll No. : AB-16030

TITLE : To prepare job as per drawing.

- OBJECTIVES :
- To familiarise with the welding equipment.
 - To aware the metal ARC welding and Gas welding.
 - To study different parameter and aspect of welding.

DRAWING:



Raw material size : (mm)

60 mm X 45 mm X 5 mm

All dia in m.m.

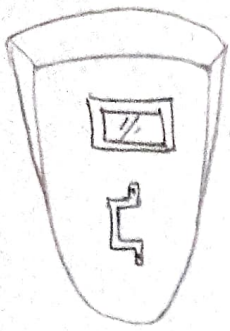
Machine/Tool/Hand Tool	Particular	Specification
1.) Source of electric and ^{arc} welding equipments.	(i) Electrode (ii) Tongue	Cast iron body
2.) Hand saw, cutting tool	(i) Bastard file (ii) Wire brush (iii) Chopping hammer (iv) 80 volt electrode	High grade carbon steel, steel blade, mild steel
3.) Electric hedder	—	Mild steel
4.) Measuring and Marking Tools	Try square	—

Sequence of operations :

- 1.) The plates are marked with chalk for correct dimensions and edges are filed upto the mark.
- 2.) One of the edges which is to be measured is marked 1mm wide and 5mm long.
- 3.) Now file the two plates.

Precautions :

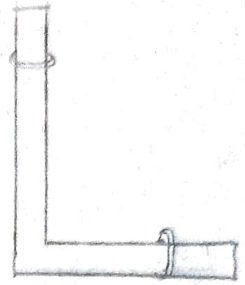
- 1.) Goggles and eye-shield must be used.
- 2.) Filler rod should be of appropriate size.
- 3.) All connections should be tight and insulated.
- 4.) The electrode should not touch the plate and should be inclined properly.
- 5.) Welding space should have properly ventilated welds.



Welding Screen



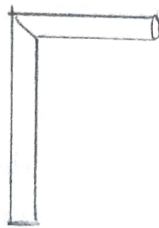
Steel Wire Brush



Try Square



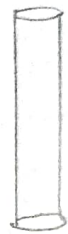
Scrubber



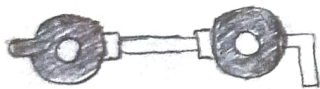
Cylindrical Key



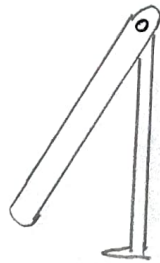
Oxygen
Hose pipe



Acetylene
pipe (Red)



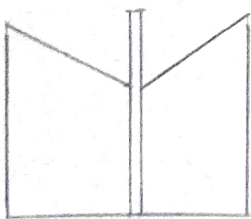
Gas Welding goggles



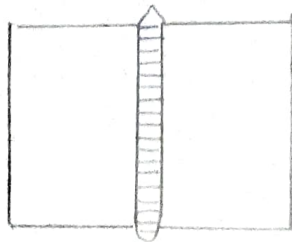
Oddley Calliper



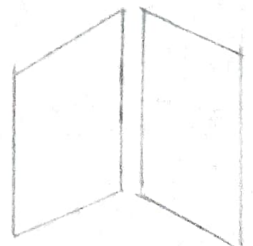
Divider



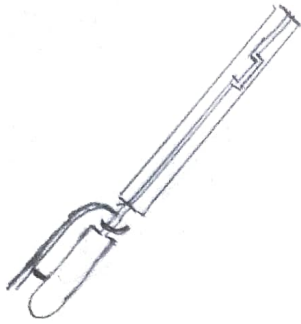
T-joint



Double-V-Butt



Corner Joint



Electric Arc Holder



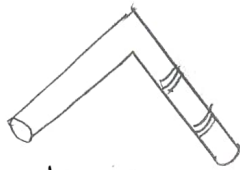
Power Cable



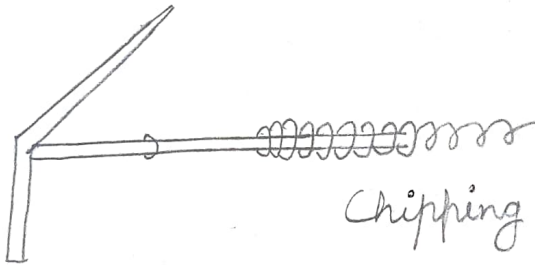
Bastard File



Y-J Pipe Joint



Y-J Pipe Corner joint



Chipping Hammer

Plate size (mm)	Electrode S.W.a.	Size (mm)	Current Range in AMPS	
			Min.	Max.
Upto 3mm	12	2.5	60	90
Upto 5mm	10	3.15	75	125
Upto 10 mm	8	4.06	140	170
Upto 15 mm	6	4.88	160	220
Upto 20 mm	4	5.89	200	275

QUESTIONARIES :

Welding 3

1. State the basic principle of Arc-Welding ?
2. Give five applications of Arc-Welding in industries ?
3. Read the following statement carefully
Decide whether these are true or false:
 - (a) Length of arc is the distance between the tip of the electrode and the work surface.
 - (b) For light work higher the current is used.
 - (c) Arc blow is the result of magnetic field in the arc stream.
 - (d) The amount of heat deposit at the two poles in DC arc welding is same.
4. How the metal being welded is protected from oxidation ?
5. How is the welding arc started ?
6. On what factor the size of electrodes depends ?
7. What is the principle; of gas welding ?
8. How different flames can be produce in welding and what are their names ?
9. What is the temperature range in gas welding and arc welding ?
10. Write any five types of welding defect and their causes ?

Q1.) State the ----- Arc-Welding?

⇒ The basic principle of arc-welding is that when two conductors in an electric circuit are brought near (2mm-4mm) such that the current continues to flow through a path of ionized particles (gaseous medium) called plasma, an electric arc is formed. This ionized gas column acts as high resistance conductor that enables more ions to flow from anode to cathode. Heat is generated as the ions strike the cathode.

Q2.) Give five applications ----- in industries?

⇒ Five application of Arc-Welding are as follows:-

(i) Construction:- Ensure strong, sustainable connections within buildings, bridges, & other infrastructure.

(ii) Employed for construction of storage devices containing air & water.

(iii) Mechanical:- Rugged applications, that involve thicker metal dimensions, arc-welding provides control & effectiveness to firmly bond heavier pieces together.

(iv) Equipments:- Properly functioning equipments require welding for tools like threshers, water systems, appliances.

(v) When toughness & compactness is required in materials & to prevent leakage.

Q3.) Read the ----- arc welding is same.

⇒ a.) False

b.) True

c.) False

d.) False.

Q4.) How the metal ----- from oxidation?

⇒ The metal being welded is protected by molten slag along with electrodes which float on metal and provide a check on its oxidation.

Q5.) How is the ----- started?

⇒ When conductors of electric circuit are brought closer & kept at a distance of 2-4mm under high potential difference, a discharge is obtained which generates high heat & leads to welding initiative.

Q6.) On what factor ----- electrodes depends?

⇒ The size of electrodes depends on:-

(i) Thickness of work piece.

(ii) Amount of current

(iii) Composition of parent material

(iv) Use of AC & DC source.

Q7.) What is the ----- gas welding?

⇒ Gas welding is done by burning a combustible gas with air or oxygen in a concentrated flow of high temperature. As with other welding flame is to

heat & melt the parent metal and filler to obtain a joint. It can weld most of the common materials.

Q8.) How different - - - - - are their names?

⇒ Different flames can be produced by varying the composition of combustible gases with O_2 . Different flames are:-

(i) Neutral flame:- It is produced when % of O_2 & acetylene is same. It is used for welding cast iron, stainless steel.

(ii) Oxidising flame:- It is produced when % of O_2 is higher. It is used for bronze welding.

(iii) Carburizing flame:- It is produced when % of acetylene is higher. It is a reducing flame.

Q9.) What is the - - - - - and arc welding?

⇒ The temperature of gas welding is $3500^\circ C$. The temperature in arc welding is $6000^\circ C$ to $7000^\circ C$.

Q10.) Write any five - - - - - their causes?

⇒ Welding defects and their causes are:-

(i) Poor fusion:- The lack of thorough and complete union between the deposited and parent metal due to faulty welding conditions.

(ii) Under-cut:- A groove melted into base metal adjacent to the toe of the weld. The reason for this are non-uniform feed of the welding rod,

improper position of electrodes or touch tip or excessive heating.

(iii) Porosity - The formation of blow holes, gas pockets or roughness on surface of weld. This is due to presence of gases in metal, moisture in flux or rust on welded edges.

(iv) slag inclusions - Presence of non-metallic substances in metal, due to contamination of base and deposited metal by oxides, high viscosity of the slag.

(v) Cracks in the weld may arise from locked up stress set up by non-uniform heating and cooling, excess sulphur or phosphorus in the weld metal or and some other cases.